

Jimmy Van Hout

vanhout@umd.edu

jimmyvanhout.com | github.com/JimmyVanHout | linkedin.com/in/JimmyVanHout

Rockville, MD

Education

University of Maryland, College Park, MD, *May 2022*

- **Bachelor of Science in Computer Science**
- Minor in Spanish Language, Culture, and Professional Contexts

Experience

Embedded Software Engineer

EmbedTek (Subcontractor), *February 2025 - Present*

- Work directly with President & CTO of a medical device manufacturer (please email me for his contact information to contact him directly) to design software and hardware of a medical device for use in hospitals and nursing homes
- Develop software for multiple ESP32 microcontrollers embedded in a medical device
- Process data from multiple sensors, control mechanical devices critical for patient health and safety, and communicate with a computer over UART or with a server over LAN
- Design hardware, including PCB, to enable software control and proper functioning of device
- Develop software on connected computer or server to store, visualize, and process data
- Please see the open-sourced component of my work on this project on GitHub at https://github.com/JimmyVanHout/temperature_pressure_sensor

Full-Stack Software Engineer

Quadrofoil (Contractor), *August 2025 - September 2025*

- Worked with client to identify and solve an issue, gather requirements, obtain product feedback from stakeholders, and deliver a product according to clients' requirements

Full-Stack Software Engineer

ScheduleAI, *June 2023 - January 2025*

- Developed a SaaS web application and artificial intelligence algorithm for automated employee scheduling

- Developed backend using Python, Flask, PostgreSQL, Amazon Cognito API; frontend using JavaScript, HTML, CSS
- Implemented server-side user authentication and authorization, including the use of OAuth 2.0, Amazon Cognito, JWTs, and cryptographic libraries
- Develop HTTP REST APIs for programmatic access of data

Software Engineer

Local fitness company (Consultant), *August 2022 - November 2022*

- Developed software to analyze and visualize customer data and resource utilization to optimize resource allocation and reduce expenses

Technical Skills

- **Programming Languages:** Python, JavaScript, C, C++, Java, C#
- **Web:** Python, Flask, JavaScript, Node.JS, HTML, CSS, React, Vue, Gunicorn, NGINX, Linux
- **Embedded:** ESP32, STM32, FreeRTOS, esp-idf, Arduino, Raspberry Pi, Linux
- **Cloud Computing/SaaS:** AWS (EC2, Cognito, Lightsail, Route53, SES, etc.), GCP, Cloudflare, Azure, Stripe
- **Security:** Authentication and authorization, OAuth 2.0, JWTs, cryptographic libraries, NIST Secure Software Development Framework (SSDF)
- **Parallel and Multithreaded Programming:** HPC cluster at UMD, MPI, OpenMP, CUDA, Hatchet, Python multiprocessing module, Java Thread library
- **Databases:** PostgreSQL, SQLite
- **Version Control:** Git, GitHub
- **Operating Systems and Shells:** Linux (Debian, Ubuntu, Raspberry Pi OS), FreeRTOS, Mac OS, Windows, Bash

Selected Technical Projects

Temperature and Pressure Sensor Program

C, Python

Read temperature and pressure from a MS5839-02BA sensor via I2C with an ESP32 microcontroller and output the data over serial. This project is an open-sourced component of medical device software currently under development.

GitHub: https://github.com/JimmyVanHout/temperature_pressure_sensor